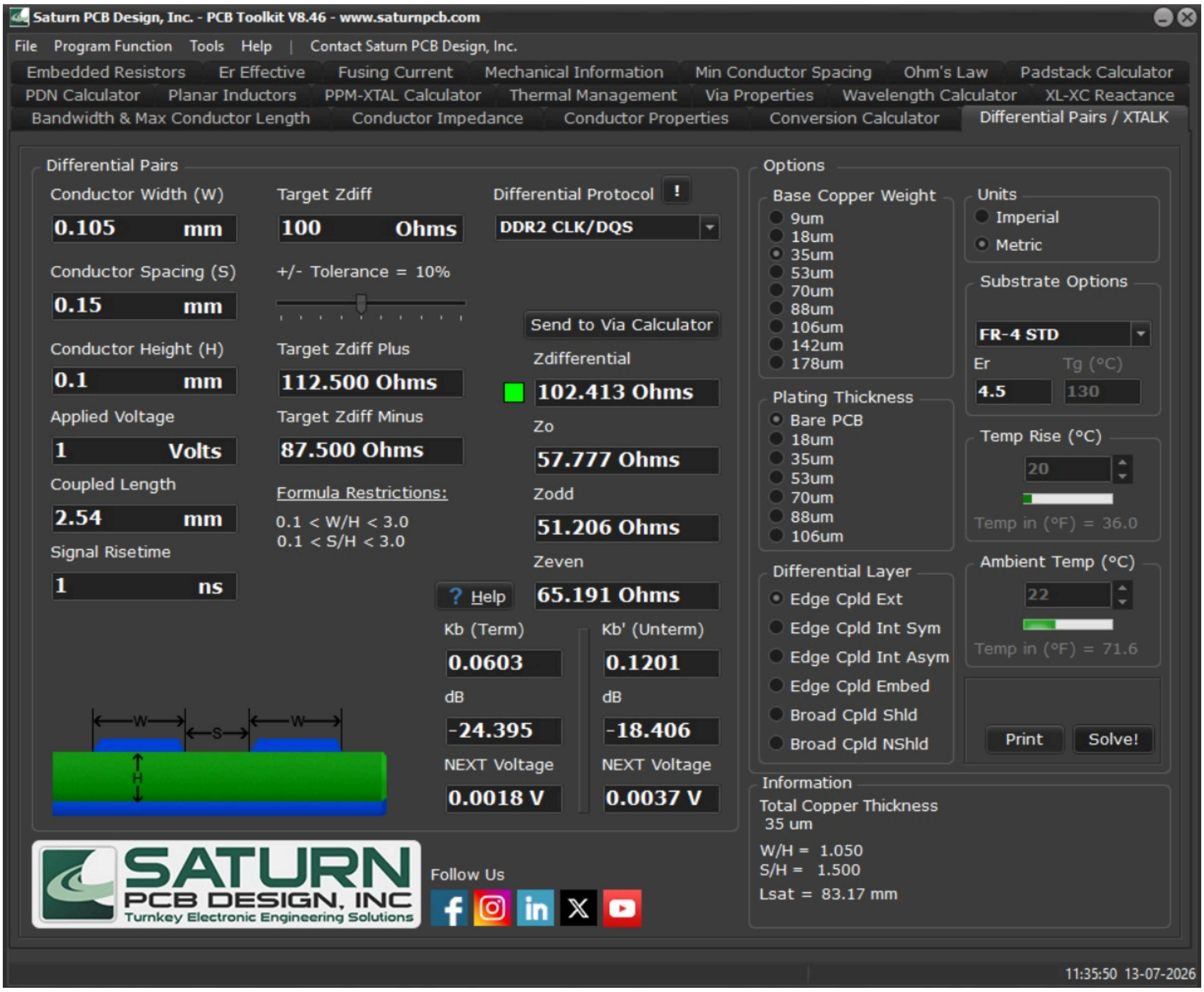
High-Speed Ethernet Differential Pair PCB Design

Candidate: Chirag Bhalla
Software: KiCad 10
Board Type: 4-Layer FR4 PCB

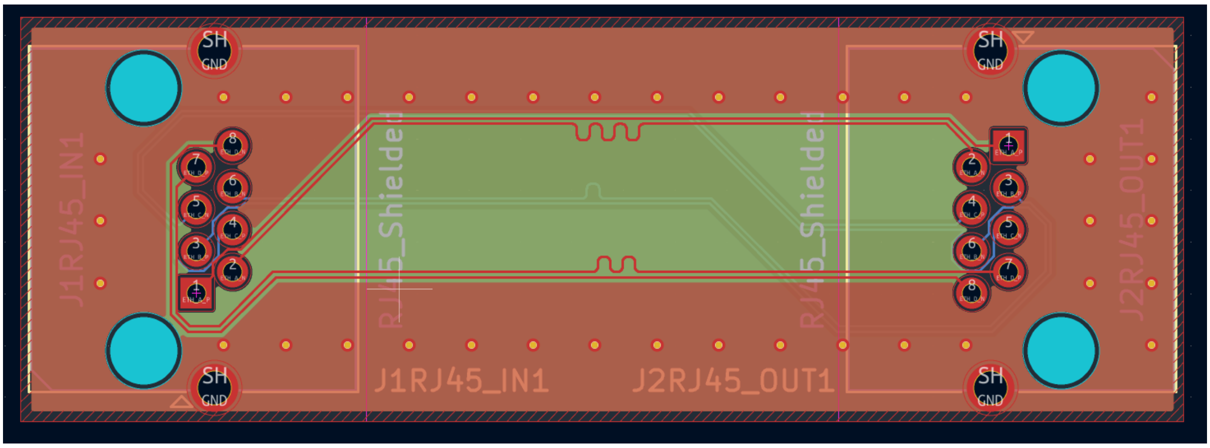
Track Geometry



A 4-layer FR-4 stackup was selected with a 0.10 mm prepreg thickness between the top signal layer and the ground reference plane. Using Saturn PCB Toolkit, the differential pair geometry was calculated as 0.105 mm trace width with 0.15 mm spacing, resulting in a calculated differential impedance of 102.4 Ω, which is within acceptable tolerance of the 100 Ω target.

Trace Width	0.105 mm
Differential Gap	0.15 mm
Copper Thickness	35 µm (1 oz)
Board Thickness	1.6 mm
Dielectric Constant (εr)	4.5
Calculated Differential Impedance	102.413Ω

High-Speed Routing



The Ethernet signals were routed as controlled differential pairs while maintaining constant trace width and spacing throughout the routing path.

To preserve signal integrity, the following routing practices were followed:

- Maintained consistent differential pair spacing.
- Used 45° bends to minimize impedance discontinuities.
- Minimized unnecessary layer transitions. The through-hole RJ45 connector pads were used where required to transition between the top and bottom routing layers without adding extra signal vias.
- Keep routing symmetrical between both connectors.
- Routed each pair together to maintain coupling.

Length Matching

To minimize **intra-pair skew**, the lengths of the positive and negative traces were matched using KiCad's Differential Pair Tune Skew tool. . This ensures that both signals in each differential pair arrive at the destination at nearly the same time.

Crosstalk Reduction

To improve signal integrity and reduce crosstalk between adjacent pairs, the following measures were implemented:

- Adequate spacing was maintained between neighbouring differential pairs.
- Continuous ground reference planes were provided on both inner layers.
- Ground stitching vias were added throughout the PCB to improve the connection between ground planes and provide a low-impedance return path.
- GND copper pours were added on both the top and bottom layers, while continuous solid ground planes were maintained on the two inner layers to provide a stable return path and improve shielding. .

These routing techniques help maintain controlled impedance while reducing noise and electromagnetic interference.

Mechanical Design

The PCB was designed as a compact Ethernet interconnect board with two shielded RJ45 connectors placed at opposite ends. This connector arrangement enables clean and symmetrical differential pair routing. while keeping the trace lengths short and maintaining a clean, symmetrical layout. The final board dimensions are **56.145 mm × 19.560 mm**, providing sufficient clearance for the connectors and routing without increasing the overall board size unnecessarily.

Since this design is intended as a passive Ethernet interconnect/evaluation board, dedicated mounting holes were not included. For a production design or enclosure-mounted application, mounting holes can be added based on the mechanical requirements of the final product.

Manufacturing Strategy

The PCB is designed using a standard 4-layer FR4 stackup with 1 oz copper on all copper layers.

F.Cu	Differential pair routing + GND copper pour
In1.Cu	Solid GND Plane
In2.Cu	Solid GND Plane

B.Cu	Differential pair routing + GND copper pour

The dedicated inner ground planes provide a stable reference for the high-speed Ethernet signals, while the top and bottom copper pours improve shielding and ground continuity.

Ground stitching vias were added across the board to connect all ground planes together, helping reduce EMI and improve return current paths.

The recommended fabrication specifications are:

Parameter	**Specification**
Material	FR4
Layers	4
Board Thickness	1.6 mm
Copper Weight	1 oz
Surface Finish	ENIG, 1 µin Gold Thickness

Before generating the manufacturing files, the following verification steps were completed:

- Design Rule Check (DRC)
- Copper zone refill
- Differential impedance verification using Saturn PCB Toolkit.
- Gerber inspection using KiCad Gerber Viewer

The final manufacturing package includes:

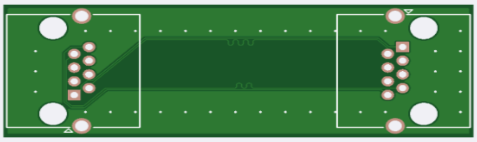
Gerber Files

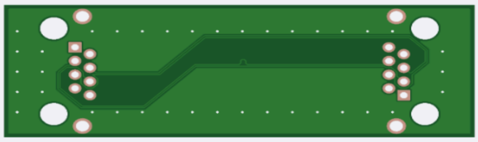
- Gerber Files
- NC Drill Files
- Board Outline
- IPC-2581 (optional)
- Pick-and-Place Files (optional)
- Bill of Materials (optional)

    All differential pairs were tuned to maintain minimal intra-pair skew. The final length differences were kept within KiCad's tuning tolerance, ensuring nearly simultaneous signal arrival for each pair.

Online PCB Quote

Re-Upload   Gerber Viewer





Detected 4 layer board of 19.56×56.42mm(0.77×2.22 inches).

Base Material	FR-4 / Flex / Aluminum / Copper Core / Rogers / PTFE Teflon
Layers	1 / 2 / 4 / High Precision PCB 6 / 8 / 10 / 12 / 14 / 16 / More
Dimensions	56.42 * 19.56 mm
PCB Qty	5
Product Type	Industrial/Consumer electronics / Aerospace / Medical

Deburring/Edge rounding	No / Yes
	For small-size PCBs, burrs are more likely to occur. If you need to remove burrs, please select 'Yes'.
Different Design	1 / 2 / 3 / 4
Delivery Format	Single PCB / Panel by Customer / Panel by JLCPCB
PCB Thickness	0.4mm / 0.6mm / 0.8mm / 1.0mm / 1.2mm / 1.6mm / 2.0mm
PCB Color	Green / Purple / Red / Yellow / Blue / White / Black
Silkscreen	White
Material Type	FR4 TG135 / FR4 TG155 / Nan Ya NP-140F / KB6164 - TG135 / Nan Ya NP-155F / KB-6165 - TG155 / S1141 TG140 / S1000H TG155
Surface Finish	HASL(with lead) / LeadFree HASL / ENIG
Gold Thickness	1 U" / 2 U"

High-spec Options

Outer Copper Weight	1 oz / 2 oz
Inner Copper Weight	0.5 oz / 1 oz / 2 oz
Specify Layer Sequence	No / Yes
Specify Stackup	No / Yes Impedance calculator >
Via Covering	Plugged / Untented / Epoxy Filled & Capped / Copper paste Filled & Capped / Tented
Via Plating Method	Not Specified / Conductive Adhesive / Horizontal Electroless Copper Plating
Min via hole size/diameter	0.3mm/(0.4/0.45mm) / 0.25mm/(0.35/0.4mm) / 0.2mm/(0.3/0.35mm) / 0.15mm/(0.25/0.3mm)
Board Outline Tolerance	±0.2mm(Regular) / ±0.1mm(Precision)
Confirm Production file	No / Yes
Mark on PCB	Remove Mark / 2D barcode (Serial Number)
Electrical Test	Flying Probe Fully Test
Gold Fingers	No / Yes
Castellated Holes	No / Yes